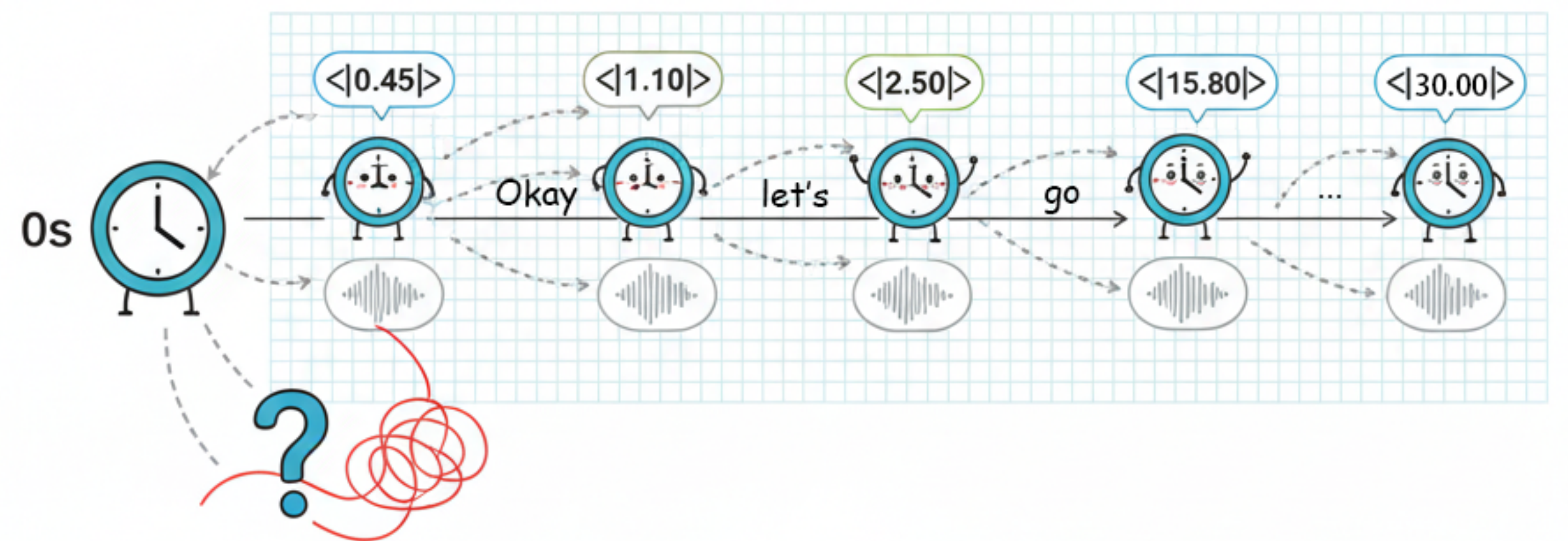


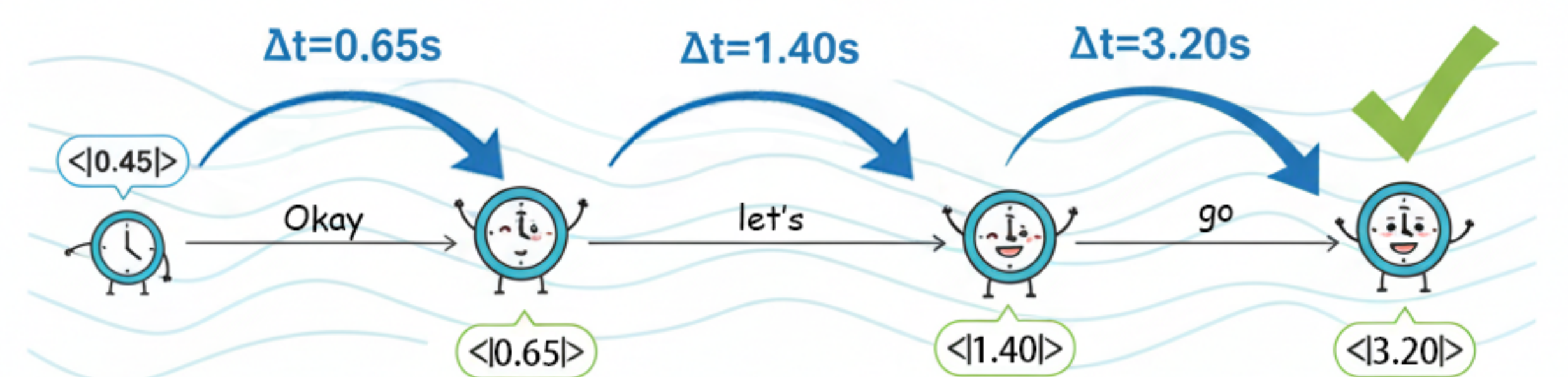
## 1. Motivation & Limitations

- **The Problem:** Speech LLMs excel at content understanding but struggle with fine-grained, word-level temporal alignment.
- **Absolute Timestamp Pitfalls:** Existing methods (e.g., Qwen2-Audio) use formats like  $\langle [0.45] \rangle$  Okay  $\langle [1.10] \rangle$  let's, leading to:
  - *Cumulative Error:* Monotonically increasing values cause error propagation.
  - *Poor Generalization:* Fails on audio longer than training duration (e.g., predicts  $> 30s$  if only trained on 30s clips).
  - *Vocabulary Explosion:* 0.01s precision over 300s audio requires  $\sim 30,000$  distinct timestamp tokens.
- **Our Solution: Relative Time Intervals & Masked Training**, transforming LLMs into “temporal-aware content understanding machines”. Instead of absolute positions, we model the **time interval** between consecutive words (e.g.,  $\langle [0.65] \rangle$  for words at 0.45s and 1.10s).
  - **Compact Vocabulary:** Only a limited range (e.g., 0.00s to 5.00s) is needed. Any length of audio can be represented cumulatively.
  - **Natural Paradigm:** Aligns perfectly with the autoregressive generation of LLMs. Avoids learning unrelated absolute values.

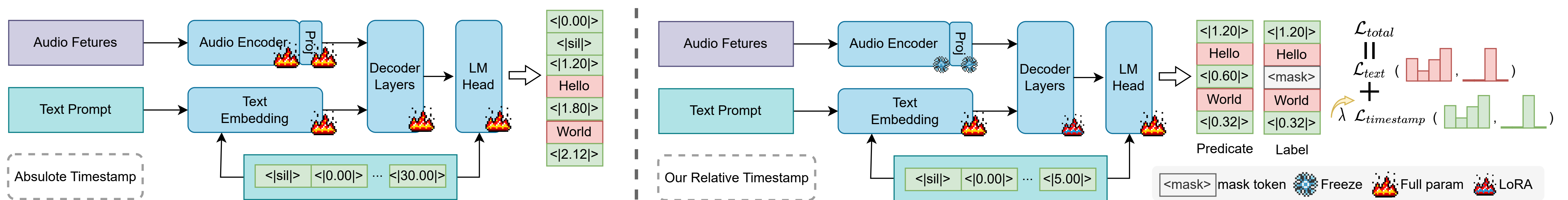
Traditional Absolute Timestmaps  
(Fixed Reference)



Our Relative Timestmaps  
(Dynamic Intervals)



## 2. Method Overview



## 3. Training Strategies & Fine-Tuning

### 1. Hybrid Parameter-Efficient Fine-Tuning:

- **Full-param:** Timestamp embedding layer & LM Head (learn new temporal vocabulary from scratch).
- **LoRA:** Decoder layers (preserve pre-trained linguistic knowledge, save compute).

### 2. Joint Temporal Alignment Loss:

$$L_{total} = L_{text} + \lambda L_{timestamp} \quad (1)$$

Dynamic weighting:  $\lambda$  starts at 1 (focus on text) and increments by 1 per epoch (shift focus to timestamps).

### 3. Timestamp Masking (Regularization):

- Randomly replaces previous timestamps with a  $\langle \text{mask} \rangle$  token (10% probability from epoch 2).
- Prevents over-relying on ground-truth history. Forces the model to *listen to the actual audio pause* to infer time, enhancing robustness against noisy annotations.

## 4. Main Results

Our method sets a new SOTA for timestamp prediction while significantly improving ASR (WER).

**Timestamp prediction (Wenet-Meeting) & ASR performance comparison.**

| Model                     | Tolerance (ms) | Precision (%) | Recall (%)   | Avg. Diff (ms) | WER (%) (Wenet) |
|---------------------------|----------------|---------------|--------------|----------------|-----------------|
| Qwen2-Audio               | 240            | 82.30         | 75.87        | 43.07          | 21.99           |
| WhisperTS                 | 240            | 43.97         | 18.79        | 102.48         | 56.66           |
| SenseVoice                | 240            | 82.82         | 81.19        | 32.47          | 7.54            |
| Abs. TS (Baseline)        | 240            | 72.41         | 68.22        | 152.17         | 11.30           |
| <b>Relative TS (Ours)</b> | <b>240</b>     | <b>91.13</b>  | <b>86.88</b> | <b>30.34</b>   | <b>5.56</b>     |

- Achieves **30.34 ms** average difference, the closest to ground truth.
- Surprisingly, adding the timestamp task **decreases WER** from 11.30% to 5.56%, proving temporal modeling forces better acoustic feature extraction.

## 6. Conclusion

We propose a novel framework addressing core challenges in precise temporal sequence understanding for Speech LLMs.

- **Beyond simple tagging:** Endows LLMs with fine-grained, aligned multi-modal spatiotemporal understanding capabilities.
- **Highly Scalable:** Compact vocabulary enables robust timestamping for *arbitrarily long audio* without degradation.
- **Mutual Enhancement:** Temporal alignment regularization actively improves standard ASR performance.

